

Akshay Borse

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years building and scaling large-scale platforms, distributed systems, and AI-powered products. Experience spans product development, platform engineering, architecture modernization, and applied AI, with systems operating at 100K+ TPS, platforms adopted across 60+ services, and products delivering \$28M+ in measured annualized impact.

EXPERIENCE

Senior Software Engineer | Microsoft | Redmond, WA

Sep 2025 to Aug 2026

- **Built the historical risk foundation for AI Agent Risk Detection**, reconstructing 90 days of multi-tenant activity to enable day-one evaluation for 237K+ autonomous agents across 13K organizations, directly applicable to Meta's AI roadmap.
- **Drove cross-service architecture across Purview IRM, Entra, and DLP** for Microsoft's AI-agent Adaptive Protection pipeline, owning end-to-end launch through Conditional Access enforcement shipped with Microsoft 365 E7.
- **Led the redesign of AI-agent ingestion for independent scale and fault isolation**, separating agent and human workloads across 28 deployments processing 135K+ security signals/day.
- **Led AI Risk Detection's sovereign-cloud architecture and rollout across GCC, GCCH and DoD**, engineering deployment infrastructure across 53 resource groups and establishing Gov-cloud Kusto telemetry.
- **Connected cross-product investigations across Microsoft's security ecosystem**, linking risk across Entra, Defender, Sentinel, and Graph, elevating customer investigation engagement 16%.
- **Designed an AI-assisted testing workflow** using agent skills and Playwright to provision test tenants, configure Copilot Studio agents with tools/policies, and validate end-to-end behavior.
- **Designed an AI-assisted debugging workflow spanning three repositories**, combining historical incident analysis with Playwright to reproduce issues, implement fixes, and create automated pull requests.

Software Engineer 2 | Amazon | Seattle, WA

Aug 2019 – Aug 2025

- **Built a 0→1 developer platform for 60+ microservices**, creating a CDK-based workflow that standardized infrastructure, authentication, observability, and CI/CD; cut service setup from 2 weeks to under 1 hour.
- **Led a monolith-to-microservices transformation at 100K+ TPS**, building 15+ services with shared REST/gRPC orchestration supporting 10M+ daily transactions while maintaining behavioral parity.
- **Designed and owned the service powering ML-predicted delivery ranges**, integrating quantile-regression models and fulfillment optimization systems to deliver \$28M in annualized business impact.
- **Replaced static delivery-risk rules with ML-driven decisioning**, deploying XGBoost on SageMaker to identify at-risk orders and trigger recovery workflows, contributing to a 7% increase in sales.
- **Launched inventory decisioning systems processing 200K+ events/day**, launching Just-In-Stock and Restock Alerting with 30-day order replay; increased restocked-item sales 13%.
- **Architected built a RAG-based product recommendation system for out-of-stock items**, surfacing relevant alternatives grounded in real-time inventory and delivery constraints, increasing sales 23%.
- **Led design and cross-functional adoption of an LLM-powered localization platform**, using Bedrock with vocabulary-constrained generation and deterministic validation across 37 email templates and 28 locales.
- **Owned the annual operational improvement roadmap across 60+ services**, driving execution across 15+ engineers and reducing Sev2 incidents 40%; led Prime Day and Black Friday readiness with zero major incidents.
- **Turned synchronized delivery for large Amazon Business orders into a first-class fulfillment capability**, driving changes across planning and fulfillment systems, eliminating 100–300 engineering hours/year.
- **Standardized observability and container infrastructure across 60+ services**, introducing shared logging and Docker libraries that cut CloudWatch spend \$250K+/month and container size 83%.

Software Engineering Intern | Liquiron | San Jose, CA

Dec 2018 to Jan 2019

- **Migrated 2 backend services to serverless**, modernizing authentication with OAuth 2.0/Passport.js and Firebase.
- **Built the API layer for a single-page application**, delivering 4 REST APIs, 10 frontend capabilities and reducing server load by 36%.

Software Engineer | Persistent Systems | Pune, India

Sep 2016 to Jul 2017

- **Built a real-time CNN classification system for images and video**, achieving 87% accuracy across 7 sentiment categories.
- **Modernized legacy web applications into a single-page architecture**, improving response time 25% and unreliable-network performance 9%; introduced an engineering feedback system adopted by 6 teams.

Software Engineering Intern | Soft Corner | Pune, India

Jun 2015 to Jun 2016

- **Improved F2FS garbage-collection efficiency 15% through Linux kernel-level performance engineering**, instrumenting filesystem behavior and benchmarking cost-benefit and greedy cleaning policies to optimize latency.

EDUCATION

- Santa Clara University, Santa Clara, CA | Master's in Computer Science and Engineering. Sep 2017 to Jun 2019
- Pune Institute of Computer Technology (PICT), India | Bachelor's in Computer Science and Engineering. Jun 2012 to Jun 2016

FEATURED PROJECTS

- **LeetDesign**: Executable system-design simulator for modeling scalability, fault tolerance, workloads, and architectural trade-offs.
- **FinanceOS**: AI Financial Intelligence Platform: Designed a hybrid deterministic + LLM architecture that computes financial facts through deterministic pipelines and uses LLM reasoning only for explanation and recommendations, keeping probabilistic models outside correctness-critical calculations.

SKILLS

- **Languages**: Java • C# • Python • C++ • C • TypeScript • JavaScript • SQL
- **Architecture & Systems**: Distributed Systems • System Design • Microservices • Event-Driven Architecture • REST APIs • gRPC • Concurrency • Fault Tolerance • High Availability • Linux
- **Cloud & Platform**: AWS • Azure • Platform Engineering • Developer Platforms • Infrastructure as Code (IaC) • CI/CD • Docker • ECS • Lambda • Step Functions • DynamoDB • Cosmos DB • Observability • SLOs
- **AI/ML & Security**: Generative AI • LLMs • RAG • AI Agents • AI Infrastructure • AI Governance • Machine Learning • NLP • XGBoost • Bedrock • SageMaker • AI Security • DLP • Entra • Purview • Playwright